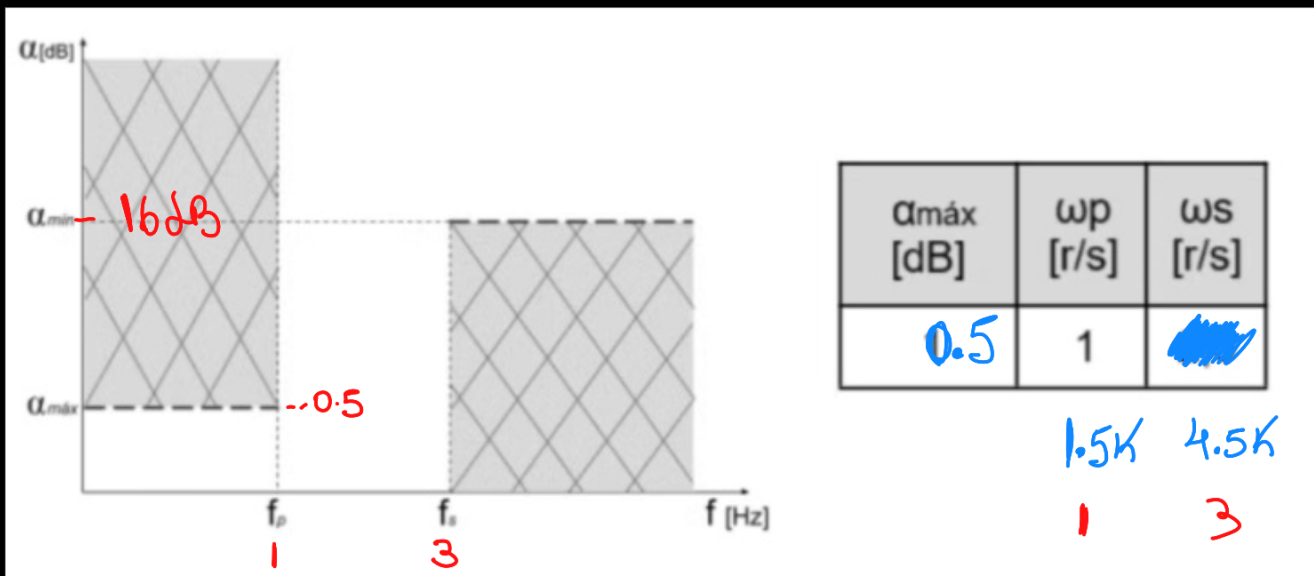




Usar un chebyshev

$$\eta = \frac{\cosh \left[\frac{\omega^{\alpha_{\min}/\omega} - 1}{\omega^{\alpha_{\max}/\omega} - 1} \right]^{1/2}}{\cosh(\omega_s)}$$

$$\eta = 2$$

$$\xi^2 = \omega^{\alpha_{\max}/\omega} - 1 = 0.122$$

$$\xi = 0.3493$$

$$|H(j\omega)|^2 = \frac{1}{1 + \xi^2 C_n^2(\omega)}$$

$$; C_2(\omega) = 2\omega^2 - 1$$

$$T(s) = \frac{c}{s^2 + s\alpha + b}$$

$$c = \frac{1}{\xi^2}$$

$$b = \sqrt{1 + \xi^2} \cdot c$$

$$\xi = 0.3493 \quad \omega_0 = \sqrt{b} = 1.231$$

$$Q = \frac{\sqrt{b}}{\alpha} = 0.863$$

$$\alpha = \sqrt{2b - 1}$$

$$c = 1.431$$

$$b = 1.5162$$

$$\alpha = 1.4256$$

$$K = \frac{c}{b} = 0.9438$$

$$T(s) = 0.944 \frac{\overbrace{1.5162}^{\omega_0^2}}{s^2 + s \underbrace{\frac{1.231}{0.863}}_{\frac{\omega_0}{Q}} + \underbrace{1.5162}_{\omega_0^2}}$$

AUkerberg - Norberg

$$K = -\frac{h_3}{h_1}$$

$$\omega_0 = \frac{1}{h_3 c}$$

$$Q = \frac{h_2}{h_3}$$

$$h_3 = \int L_z = 1$$

$$K = -\frac{1}{h_1}$$

$$\omega_0 = \frac{1}{c}$$

$$Q = h_2$$